Task 4: Solution Decision Framework & Technical Proposal

Week 3 Assessment | Yoho AI Advisory | Decision Framework & Governance

1. Solution Decision Framework (5 Business Problems)

Problem / Use Case	Input Type	Output Type	Constraints (Latency, Cost, Explainability)	Tool Choice	Human Review & Governance
1. GST Calculation on Invoices	Tabular / Numeric invoice line items	Exact tax amount ($ or ■)	• Latency: <10ms • Cost: Zero • Explainability: 100% strict compliance	**Deterministic Code / Rule Engine (Python / SQL)**	**AI SHOULD NOT BE USED.** Zero tolerance for arithmetic hallucination or tax calculation drift.
2. Current Leave Balance of Employee	Employee ID / Date query	Integer balance value	• Latency: <50ms • Cost: Zero • Explainability: 100% audit log trail	**Relational Database (SQL query) / HRIS API**	**AI SHOULD NOT BE USED.** Transactional record lookup must be exact and auditable.
3. Ranking Resumes for a Role	Unstructured text (PDF/DOCX resumes + Job Description)	Ranked candidate list with match scores	• Latency: Async/Minutes • Cost: Moderate • Explainability: Critical (Anti-bias & legal)	**Dense Embeddings + Hybrid Semantic Search + LightGBM**	**MANDATORY HUMAN REVIEW.** AI produces shortlist; human recruiter must make final shortlisting/hiring decisions to avoid bias.
4. Contract Q&A; (60-page PDF)	Complex legal PDF document + User natural language queries	Specific extracted clauses with citation anchors	• Latency: 1-3 sec • Cost: Low ($0.002/query) • Explainability: High (Must cite page & clause)	**Chunked RAG (Vector DB + Gemini Flash + Reranker)**	**HUMAN IN THE LOOP.** Legal counsel reviews high-stakes liability and termination obligations before signing.
5. Forecasting Next Month's Sales	Historical time-series sales, seasonality & market indicators	Continuous numeric demand forecast with confidence intervals	• Latency: Batch / Daily • Cost: Low • Explainability: High (Feature attribution)	**Time-Series ML (LightGBM / Prophet / ARIMA)**	**HUMAN REVIEW ON ANOMALIES.** Operations & finance leadership reviews forecasts differing by >15% from trend.

2. Technical Proposal: Contract Q&A; (60-Page Contract)

Client: Yoho Stakeholders | **Project:** Enterprise Legal Clause Intelligence System

A. Recommended Approach

We recommend an **Agentic Chunked RAG (Retrieval-Augmented Generation) Pipeline with Strict Citation Grounding**. A 60-page contract (~30,000 words / 40,000 tokens) contains interconnected legal covenants, jurisdiction clauses, and termination rights. The pipeline ingests the PDF using OCR-aware layout extraction (PyMuPDF), chunks by structural legal clauses (500 tokens with 100-token overlap), embeds via text-embedding-004 into an in-memory vector store (ChromaDB / FAISS), and retrieves top-5 clause candidates. A Cross-Encoder reranker selects the top-3 most relevant clauses, which are passed to **Gemini 2.5 Flash** with a temperature of 0.0 and a strict system instruction requiring explicit page and paragraph citations for every assertion.

B. Data Requirements

1. Document Ingestion: Clean digital PDF contracts (including scanned documents parsed via OCR).
2. Benchmark Evaluation Set: 50 expert-annotated legal queries paired with ground-truth clause citations and human-verified answers.
3. Legal Ontology / Metadata: Document metadata tags (effective date, counterparties, governing law, clause headers).

C. Evaluation Metric & Quantitative Target

Primary Metric: Citation Grounding Accuracy & Hallucination Rate evaluated via RAGAS framework (Faithfulness & Answer Relevance).
Quantitative Targets:
• **Citation Precision:** ≥ 98% (every claim must point to the exact page and section).
• **Factual Faithfulness (Zero Hallucination):** 100% verified against source text.
• **Retrieval Recall@3:** ≥ 95% across critical clauses (indemnity, breach, termination, liability caps).

D. Estimated Build Time & Phased Roadmap

Total Timeline: 3 Weeks (MVP to Production)
• **Week 1: Document Parsing & Ingestion:** Legal PDF parsing, hierarchical chunking, vector indexing, and citation mapping.
• **Week 2: Retrieval & LLM Pipeline:** Hybrid search (BM25 + Dense embeddings), reranker integration, and prompt engineering with guardrails.
• **Week 3: Evaluation & Human-in-the-Loop UI:** Automated testing against 50 annotated test cases, latency tuning, and side-by-side citation review UI.

E. Key Trade-offs & Architecture Decisions

• **Chunked RAG vs. Full-Document Direct Context:** While Gemini supports a 1M+ token context window, stuffing the full 60 pages into every prompt increases cost 40× ($0.03 vs. $0.0008 per query) and induces latency (3-5s vs. <1.2s). Chunked RAG delivers sub-second answers with surgical precision.
• **Speed & Cost vs. Ultra-Heavy Reasoning:** Gemini 2.5 Flash with deterministic retrieval provides legal-grade accuracy at 1/10th the cost of heavyweight frontier reasoning models.

3. 3-Minute Presentation Pitch & 'Why Not Just Ask ChatGPT?'

The 3-Minute Pitch (Executive Summary):
'Legal review of a 60-page contract takes an attorney 3 to 4 hours. Our proposed Contract Q&A; system cuts clause discovery to under 2 seconds while guaranteeing 98%+ citation precision. By pairing clause-level semantic retrieval with deterministic temperature controls, we deliver an automated legal assistant that highlights the exact clause, page number, and binding language for human counsel approval.'

Defending: 'Why Not Just Ask ChatGPT?'
1. Data Privacy & Enterprise IP Leakage: Pasting proprietary 60-page client contracts into public, web-based consumer ChatGPT risks exposing confidential M&A;, NDA, and liability terms to public training corpora. Our system uses enterprise-grade zero-data-retention APIs.
2. Hallucination & Lack of Audit Trails: ChatGPT can confidently hallucinate non-existent termination clauses or conflate boilerplate language across different sections. Our system forces verifiable citation grounding (Page X, Section Y), allowing instant legal verification.
3. 'Lost in the Middle' Attention Decay: In 60-page raw text dumps, general LLMs often miss nuanced carve-outs buried in appendix disclosures. Chunked hybrid retrieval ensures critical indemnity qualifiers are never overlooked.